AP BIOLOGY TOPIC 2.3

Plasma Membrane: The Fluid Mosaic Model & Amphipathic Architecture

UNIT 2: CELL STRUCTURE AND FUNCTION (10–13% EXAM WEIGHT)

ENE-2.A | ENE-2.B | FLUID MOSAIC MODEL | AMPHIPATHIC

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 2.3 (PLASMA MEMBRANE: THE FLUID MOSAIC MODEL & AMPHIPATHIC ARCHITECTURE)

CED Core Vocabulary

Amphipathic Molecules — Phospholipids possess both a hydrophilic (water-loving, polar) region and a hydrophobic (water-fearing, nonpolar) region: Hydrophilic Head: Glycerol + negatively...

Spontaneous Bilayer Formation — In aqueous environments, phospholipids spontaneously self-assemble into a bilayer (driven by entropy/hydrophobic exclusion of water), creating a stable,...

Cell Signaling & Adhesion — Tissue sorting during embryonic development and intercellular communication.

R — Receptors (Signal Transduction) — Specific binding sites for hormones/neurotransmitters (ligands).

Hydrophobic Tails — Two long nonpolar fatty acid hydrocarbon chains → point inward away from water, forming a nonpolar interior core held together by weak van der Waals /...

Key Biological Roles — Cell-to-Cell Recognition & Immune Response: Distinguish "self" from foreign "non-self" cells (e.g., MHC — Major Histocompatibility Complex glycoproteins in...

T — Transport — Hydrophilic channels, carriers, and ATP-driven pumps.

A — Attachment / Anchorage — Link intracellular cytoskeleton (microfilaments) to extracellular matrix (ECM) proteins (e.g., integrins, fibronectin, collagen) for structural stability.

§2. PHOSPHOLIPID BILAYER: AMPHIPATHIC SELF-ASSEMBLY

Core Framework

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Phospholipid Bilayer ➔ **Amphipathic Molecules** ➔ **Spontaneous Bilayer Formation**

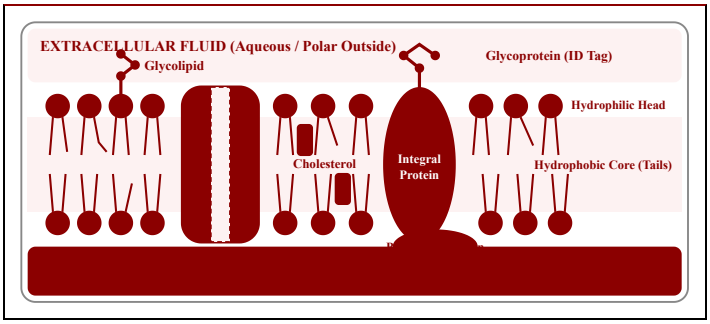
- **Amphipathic Molecules:** Phospholipids possess both a **hydrophilic (water-loving, polar)** region and a **hydrophobic (water-fearing, nonpolar)** region:
  - **Hydrophilic Head:** Glycerol + negatively charged **phosphate group (** $-PO_4^{2-}$ **)** + choline → faces outward toward aqueous **extracellular fluid** and inward toward aqueous **cytosol**.
  - **Hydrophobic Tails:** Two long nonpolar **fatty acid hydrocarbon chains** → point inward away from water, forming a **nonpolar interior core** held together by weak van der Waals / hydrophobic interactions.
- **Spontaneous Bilayer Formation:** In aqueous environments, phospholipids spontaneously self-assemble into a **bilayer** (driven by entropy/hydrophobic exclusion of water), creating a stable, **selectively permeable barrier** separating internal vs. external environments.

[COMMON AP EXAM FRQ TRAP] **The plasma membrane is DYNAMIC and FLUID, NOT rigid or static!** Phospholipids and many proteins drift laterally (~2 μm/sec) within their leaflet because they are held by weak hydrophobic interactions, not covalent bonds.

[EXAMPLE & BIOLOGICAL APPLICATION] **Dynamic Membrane Flexibility:** Enables **white blood cells (leukocytes)** to deform and squeeze through narrow capillary walls (diapedesis), self-seal after microneedle puncture, and form/fuse vesicles during **endocytosis & exocytosis**.

§3. FLUID MOSAIC MODEL VISUALIZED

Labeled Cross-Section



§5. MEMBRANE PROTEINS: INTEGRAL VS. PERIPHERAL

Mosaic of Functions

- Membrane proteins can be **hydrophilic, hydrophobic, or amphipathic** depending on their amino acid R-groups:

PROTEIN CLASS	LOCATION & R-GROUP POLARITY	STRUCTURAL & FUNCTIONAL ROLES
Integral (Transmembrane) Proteins	Embedded through the lipid bilayer; **nonpolar/hydrophobic R-groups** (often α-helices) anchor in the fatty acid core, while **polar/charged R-groups** protrude into water on both sides	**Channels, carrier proteins, active pumps** (Na^+/K^+ pump), signal transduction **receptors** (GPCRs, RTKs)
Peripheral Proteins	Loosely bound to the **hydrophilic surface** of the membrane (often attached to integral proteins or polar lipid heads); rich in **polar/charged R-groups**	Cytoskeletal anchoring, intracellular signaling cascades, enzymes, structural support

- **6 Major Functions of Membrane Proteins (Mnemonic: *"TRACIE"*):**
  - **T — Transport:** Hydrophilic channels, carriers, and ATP-driven pumps.
  - **R — Receptors (Signal Transduction):** Specific binding sites for hormones/neurotransmitters (ligands).
  - **A — Attachment / Anchorage:** Link intracellular **cytoskeleton (microfilaments)** to **extracellular matrix (ECM)** proteins (e.g., integrins, fibronectin, collagen) for structural stability.
  - **C — Cell-Cell Recognition:** Glycoproteins serve as molecular identification tags.
  - **I — Intercellular Joining:** Form tight junctions, desmosomes, and gap junctions between adjacent cells.
  - **E — Enzymatic Activity:** Catalyze sequential steps of a metabolic pathway right at the membrane surface.

[AP EXAM FRQ SCORING STRATEGY] **Amino Acid Mutation FRQ**

Predictions: If a mutation replaces a **nonpolar amino acid** (e.g., Leucine/Valine) in the transmembrane domain of an integral protein with a **charged/polar amino acid** (e.g., Glutamate/Lysine), the protein **cannot fold or embed properly** into the hydrophobic fatty acid tail core!

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§4. GLYCOPROTEINS & GLYCOLIPIDS (EXTRACELLULAR SURFACE)

Cell Recognition

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Glycoproteins & Glycolipids (Extracellular Surface) ➔ **exterior (extracellular) face** ➔ **Key Biological Roles**

- Found exclusively on the **exterior (extracellular) face** of the plasma membrane (forming the *glycocalyx*):
  - **Glycoprotein = Membrane Protein + covalently bonded Carbohydrate chain** (oligosaccharide).
  - **Glycolipid = Phospholipid + covalently bonded Carbohydrate chain**.
- **Key Biological Roles:**
  - **Cell-to-Cell Recognition & Immune Response:** Distinguish "self" from foreign "non-self" cells (e.g., **MHC — Major Histocompatibility Complex** glycoproteins in vertebrate immunity; **A, B, O blood group antigens** on red blood cells).
  - **Cell Signaling & Adhesion:** Tissue sorting during embryonic development and intercellular communication.

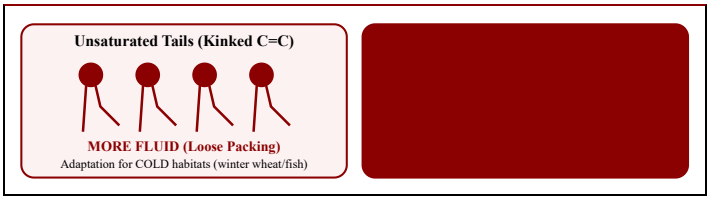
[CORE AXIOM / REMEMBER] **Carbohydrates on the membrane = Cell ID**

Tags! Always oriented toward the *outside (extracellular side)* of the plasma membrane.

§6. MEMBRANE FLUIDITY CONTROL: TAILS & CHOLESTEROL

Homeoviscous Adaptation

- **1. Saturated vs. Unsaturated Fatty Acid Tails (All Cells):**
  - **Unsaturated Hydrocarbon Tails:** Have one or more ***cis* double bonds (C=C)** that introduce **kinks** → prevent tight packing → **INCREASE membrane fluidity** (prevent freezing in cold environments).
  - **Saturated Hydrocarbon Tails:** Only single bonds (C–C); straight chains pack tightly together via van der Waals attractions → **DECREASE fluidity** (more viscous; adaptation for hot environments).
- **2. Cholesterol (Temperature Fluidity Buffer in Animal Cell Membranes):**
  - Steroid lipid wedged between phospholipid tails in **animal** plasma membranes (plants use phytosterols; prokaryotes lack cholesterol).
  - **At High Temperatures (e.g., 37°C+):** Restrains excessive phospholipid movement → **REDUCES fluidity** and stabilizes the membrane against melting/leakiness.
  - **At Low Temperatures:** Hinders close packing of fatty acid tails → **PREVENTS solidification (freezing)** and maintains fluidity.



[CORE AXIOM / REMEMBER] **Cholesterol = Temperature Fluidity Buffer!**

- **High Temp:** Restrains phospholipid movement → **decreases fluidity** (prevents melting).
- **Low Temp:** Disrupts tight tail packing → **prevents solidification** (keeps membrane fluid).

§7. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 2.3 [MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]

Topic 2.3: Plasma Membrane: The Fluid Mosaic Model & Amphipathic Architecture ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigation 4 (Diffusion & Osmosis): Dialysis tubing & potato core mass change (% ΔMass = [(Final − Initial) / Initial] × 100) across sucrose molarities (0.0 M to 1.0 M). IV: Extracellular sucrose molarity (M); DV: % change in mass; Isotonic point: x-intercept where % ΔMass = 0% (no net water movement). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Graph Rules: Total Water Potential: $\Psi = \Psi_p + \Psi_s$. Solute Potential: $\Psi_s = -iCRT$ (where i = ionization constant [1 for sucrose/glucose, 2 for NaCl], C = molarity in mol/L, R = 0.0831 L·bar/mol·K, T = K = °C + 273). Surface-Area-to-Volume Ratio (SA:V): Cube SA = $6s^2$, V = s^3 (SA:V = 6/s). Larger SA:V accelerates diffusion per unit volume. **Error Bar Rule (±2 SE_x̄):** If the ±2 $SE_{\bar{x}}$ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Physiological Disruption Case Study: Cystic Fibrosis results from a deletion (ΔF508) in the CFTR chloride channel protein, preventing misfolded channels from reaching the apical plasma membrane. Impaired Cl^- efflux prevents osmotic water movement into mucus, producing thick, dehydrated mucus in lungs and pancreatic ducts. Ouabain/Digoxin inhibits the Na^+/K^+-ATPase pump, collapsing the electrochemical gradient required for secondary active cotransport.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 2.3 to prove why the claim holds.

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